

AI-Assisted Plasma Confinement Optimization in a Dual-Field Toroidal System

Abstract

This study presents a conceptual and simulation-based plasma confinement approach utilizing dual magnetic fields within a toroidal geometry. An adaptive control framework is introduced to regulate plasma stability and reduce wall interaction. The results demonstrate that dynamic field adjustment improves confinement behavior while reducing magnetic intensity requirements. Detailed implementation parameters are intentionally abstracted.

Introduction

Fusion systems such as tokamaks require high magnetic fields and face stability challenges. This work proposes a control-oriented approach focusing on adaptive optimization rather than static magnetic confinement.

System Model

The system is based on toroidal geometry with interacting variables including plasma density, thermal state, and radial behavior. Magnetic confinement is achieved using dual-field structures producing helical stabilization. Exact parameterization is intentionally not disclosed.

Control System

An adaptive feedback mechanism dynamically adjusts system variables based on plasma conditions. The controller maintains stability while optimizing efficiency. Specific coefficients and tuning parameters are intentionally omitted.

Results

Simulation results show stable plasma conditions ($\sim 1000^\circ\text{C}$), reduced wall interaction, and decreasing magnetic field requirements over time. Results are presented as representative ranges.

Discussion

The findings suggest adaptive control can improve plasma confinement efficiency. Real-world implementation will require further work in materials, magnetic systems, and diagnostics. Some implementation details are intentionally excluded.

Conclusion

This study demonstrates that adaptive control combined with dual-field toroidal geometry can achieve stable plasma confinement with improved efficiency. The approach highlights the importance of control strategies in next-generation systems.